NAVAL ARCHITECTURE
STABILITY AT DOCKING

1 Just before entering dry-dock a ship 5000 tonne displacement floats at draughts of 2.7 m forward and 4.2 m aft. The length between perpendiculars is 150 m and the water has a density of 1025 kg/m^3. Using the given hydrostatic data, find:

a. The thrust on the heel of the stern frame, which is at the aft perpendicular, when the ship just hits the docking blocks, and
b. The metacentric height at the instant of setting on the blocks.

Hydrostatic Data:

KG = 8.5 m
KM_T = 9.3 m
MCT1cm = 105 MN
LCF = 2.7 m aft of midship

[2.18 MN; 0.39 m]

2 A ship 6000 tonne displacement enters dry-dock with a draught of 4.5 m aft. If the maximum allowable thrust on the stern frame is set at 3.0 MN, determine:

a. the draught forward
b. the virtual metacentric height

at the time of docking, given the following hydrostatic data:

- LBP = 100 m
- LCF = 2.5 m aft
- KM_T = 9.0 m
- KM_L = 105 m
- KG = 5.5 m

3 Describe the process of docking a ship given reasons for the actions taken.